1 Introduction. Linear models
1.1 Artificial intelligence, machine learning, and deep learning

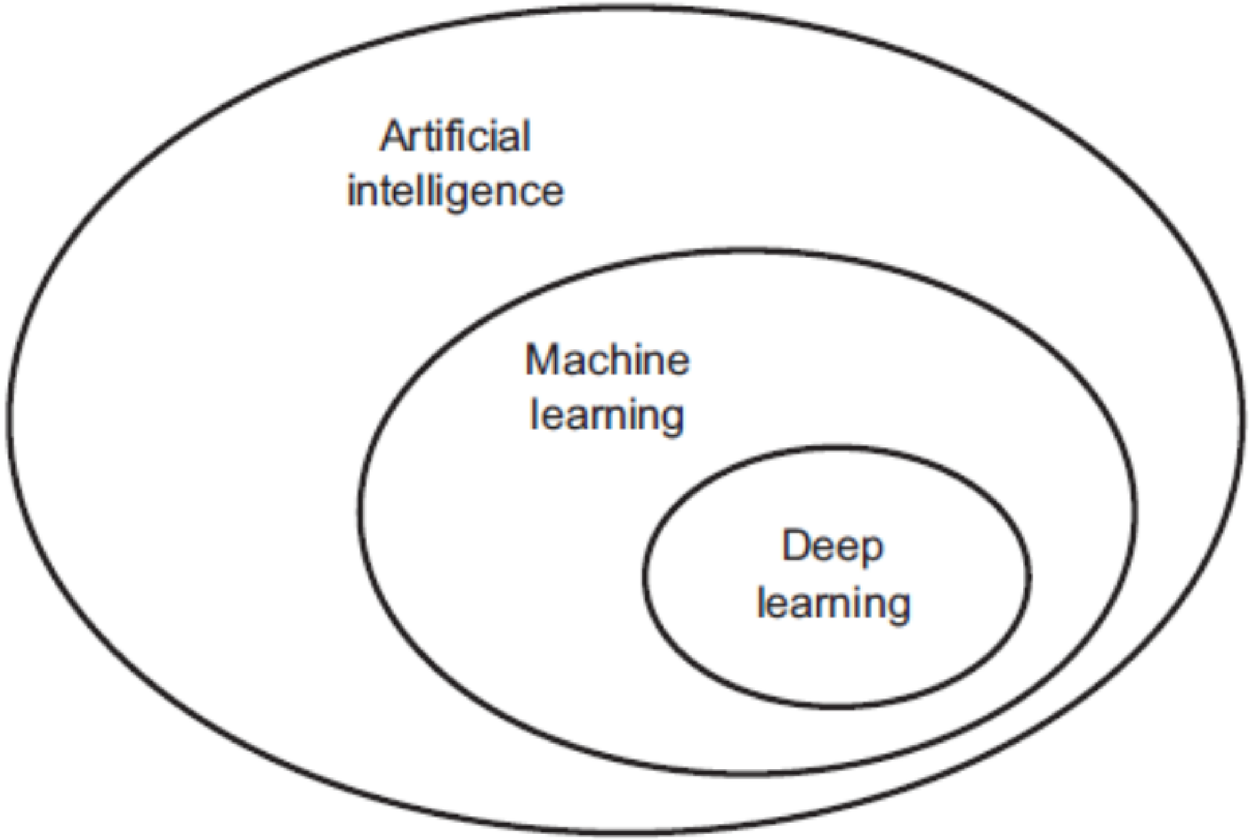


Figure 1: Artificial intelligence, machine learning, and deep learning.

1.1.1 Artificial intelligence

- *the effort to automate intellectual tasks normally performed by humans*
- *symbolic AI*: programmers handcraft a sufficiently large set of explicit rules for manipulating knowledge
- peak popularity during the *expert systems* boom of the 1980s
- symbolic AI – suitable to solve well-defined, logical problems, but intractable to figure out explicit rules for solving more complex, fuzzy problems

1.1.2 Machine learning

- rather than programmers crafting data processing rules by hand, could a computer automatically learn these rules by looking at data?

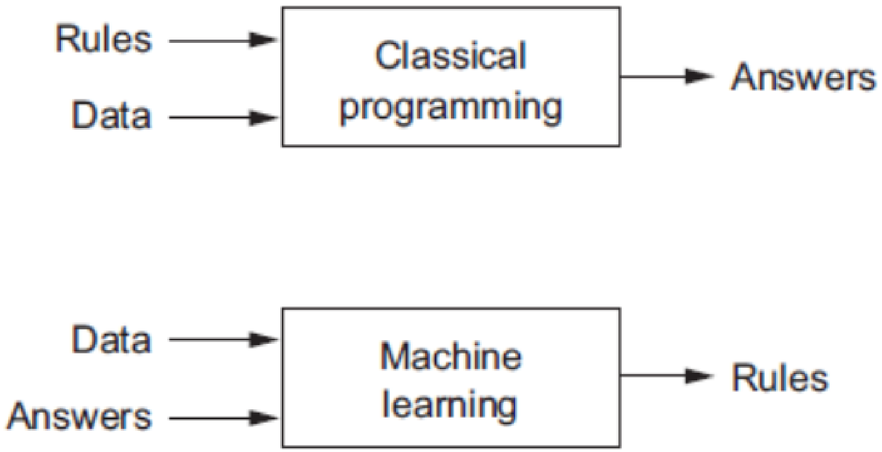


Figure 2: Machine learning: a new programming paradigm.

- a machine learning system is *trained* rather than explicitly programmed
- although machine learning only started to flourish in the 1990s, it has quickly become the most popular and most successful subfield of AI

1.1.3 Learning representations from data

- machine learning discovers rules to execute a data processing task, given examples of what's expected
- so, to do machine learning, we need three things:
  - *Input data points*
  - *Examples of the expected output*
  - *A way to measure whether the algorithm is doing a good job*
- the central problem in machine learning and deep learning is to *meaningfully transform data*: in other words, to learn useful *representations* of the input data at hand – representations that get us closer to the expected output
- representation – a different way to look at data – to *represent* or *encode* data

1.1.3 Learning representations from data

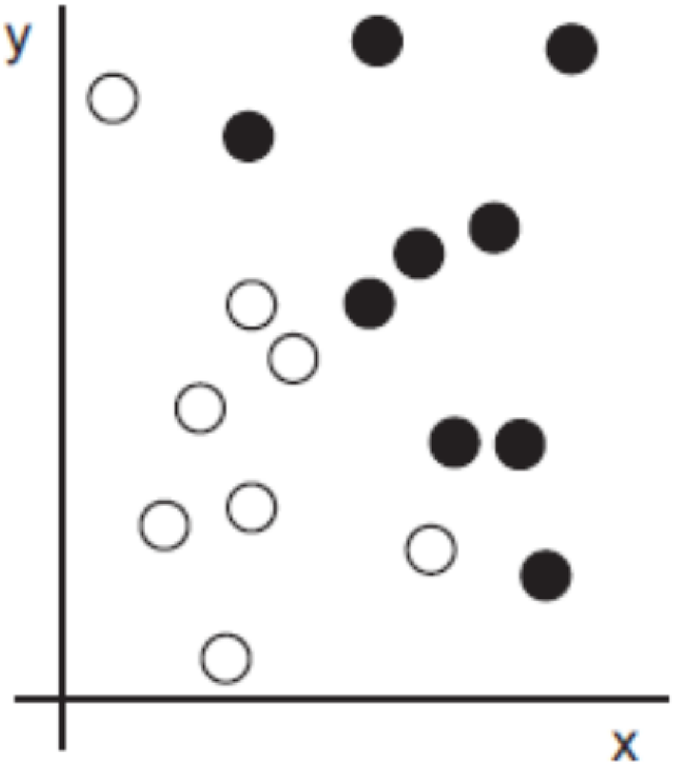


Figure 3: Some sample data.

- inputs?
- expected outputs?
- a way to measure whether the algorithm is doing a good job?

1.1.3 Learning representations from data

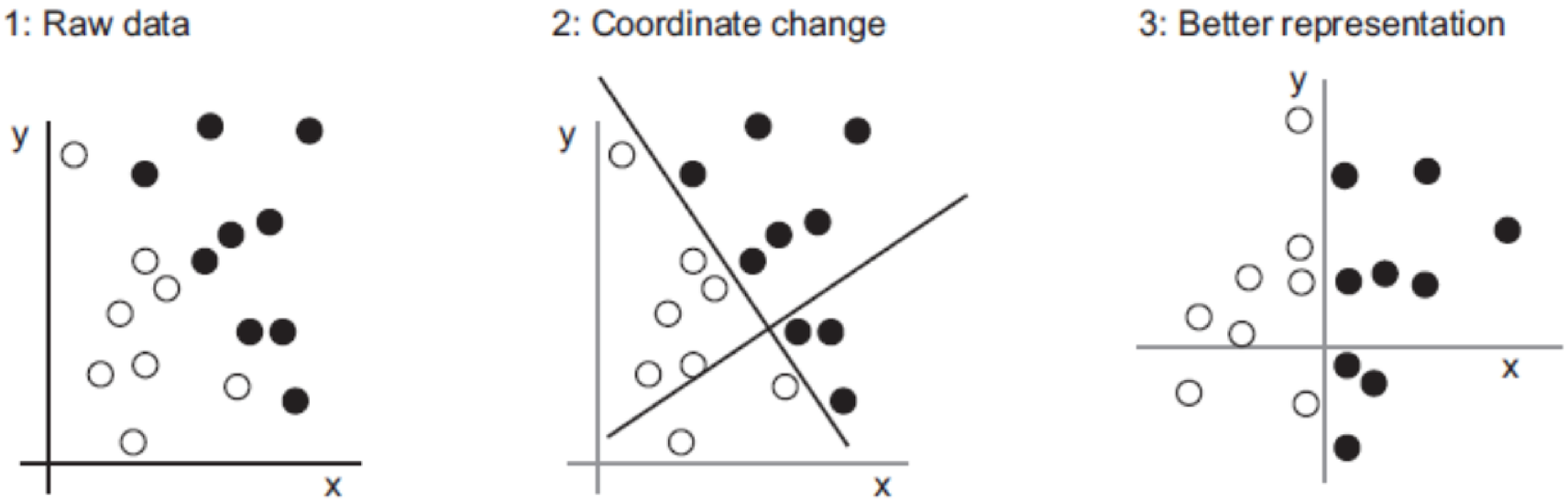


Figure 4: Coordinate change.

- with this representation, the black/white classification problem can be expressed as a simple rule: “Black points are such that $x > 0$”, or “White points are such that $x < 0$”

1.1.3 Learning representations from data

- if we tried systematically searching for different possible coordinate changes, and used as feedback the percentage of points being correctly classified – machine learning
- *learning*, in the context of machine learning, describes an automatic search process for better representations
- machine learning algorithms aren't creative in finding transformations; they're merely searching through a predefined set of operations, called a *hypothesis space*
- two main types of problems:
  - *classification*, where the goal is to predict a single discrete label for an input data point
  - *regression*, which consists of predicting a continuous value instead of a discrete label

1.1.4 Four branches of machine learning
Supervised learning

- it consists of learning to map input data to known targets (also called *annotations*), given a set of examples (often annotated by humans)
- generally, almost all applications of machine learning that are in the spotlight these days belong in this category, such as speech recognition, image classification, and language translation
- although supervised learning mostly consists of classification and regression, there are more exotic variants as well, including the following:
  - *Sequence generation*—Given a picture, predict a caption describing it.
  - *Syntax tree prediction*—Given a sentence, predict its decomposition into a syntax tree.
  - *Object detection*—Given a picture, draw a bounding box around certain objects inside the picture. This can also be expressed as a classification problem (given many candidate bounding boxes, classify the contents of each one) or as a joint classification and regression problem, where the bounding box coordinates are predicted via vector regression.
  - *Image segmentation*—Given a picture, draw a pixel-level mask on a specific object.

Unsupervised learning

- this branch of machine learning consists of finding interesting transformations of the input data without the help of any targets, for the purposes of data visualization, data compression, or data denoising, or to better understand the correlations present in the data at hand
- unsupervised learning is at the core of data analytics, and it's often a necessary step in better understanding a dataset before attempting to solve a supervised learning problem
- *dimensionality reduction* and *clustering* are well-known categories of unsupervised learning

Self-supervised learning

- this is a specific instance of supervised learning, but it's different enough that it deserves its own category
- self-supervised learning is supervised learning without human-annotated labels
- there are still labels involved (because the learning has to be supervised by something), but they're generated from the input data, typically using a heuristic algorithm
- for instance, *autoencoders* are a well-known instance of self-supervised learning, where the generated targets are the input, unmodified
- in the same way, trying to predict the next frame in a video, given past frames, or the next word in a text, given previous words, are instances of self-supervised learning (*temporally supervised learning*, in this case: supervision comes from future input data)

Reinforcement learning

- long overlooked, this branch of machine learning recently started to get a lot of attention after Google DeepMind successfully applied it to learning to play Atari games (and, later, learning to play Go at the highest level)
- in reinforcement learning, an *agent* receives information about its environment and learns to choose actions that will maximize some reward
- for instance, a neural network that “looks” at a video game screen and outputs game actions in order to maximize its score can be trained via reinforcement learning
- currently, reinforcement learning is mostly a research area and hasn’t yet had significant practical successes beyond games
- in time, however, we expect to see reinforcement learning take over an increasingly large range of real-world applications: self-driving cars, robotics, resource management, education, and so on